Unless explicitly requested by a problem, do not include sketches as part of your proof. You are free to use the result from any problem on this (or previous) assignment as a part of your solution to a different problem even if you have not solved the former problem.

Problem 1 (2pts)**.** Let G be any graph. Prove that

$$|E(G)| \geqslant \binom{\chi(G)}{2}.$$

Solution. Let $t = \chi(G)$. We know that we can partition $V(G) = A_1 \sqcup \cdots \sqcup A_t$ into t independent sets. Now we claim that for any pair $i \neq j \in \{1, \ldots, t\}$, there is an edge going between A_i and A_j. Indeed, if there was no such edge, then also $A_i \sqcup A_j$ is an independent set, which means that we can partition $V(G)$ into $t-1$ independent sets. This contradicts the fact that $t = \chi(G)$ is the smallest integer for which such decomposition is possible.

Hence, there is an edge between any pair of these independent sets. Moreover, since the sets are disjoint, this gives $\binom{t}{2}$ edges in G, as desired. □

We give another solution using critical graphs.

Solution. Let $t = \chi(G)$. We know from class that G contains a t-critical graph H. Since H is t-critical, we have $\delta(H) \geqslant t-1$, which also implies $|V(G)| \geqslant t$. Thus, by the handshaking lemma, we have

$$|E(G)| \geqslant |E(H)| = \frac{1}{2} \sum_{v \in V(H)} \deg_H v \geqslant \frac{1}{2} \sum_{v \in V(H)} (t-1) = \frac{1}{2}|V(H)|(t-1) \geqslant \frac{1}{2}t(t-1) = \binom{t}{2},$$

as desired. □

Problem 2 (1pt)**.** Let $G = (V, E)$ be any graph and consider a (not necessarily proper) edge-coloring $f\colon E \to \{\text{red}, \text{blue}\}$). Let G_r and be the graph formed by the red edges and let G_b be the graph formed by the blue edges (formally, $G_r = \big(V, f^{-1}(\text{red})\big)$ and $G_b = \big(V, f^{-1}(\text{blue})\big)$). Prove that

$$\chi(G_r) \cdot \chi(G_b) \geqslant \chi(G).$$

Hint: this generalizes the theorem $\chi(G) \cdot \chi(\overline{G}) \geqslant |V(G)|$ from Lecture 20.

Solution. Put $t_r = \chi(G_r)$ and $t_b = \chi(G_b)$. Let $f_r\colon V \to \{1, \ldots, t_r\}$ and $f_b\colon V \to \{1, \ldots, t_b\}$ denote proper colorings for G_r and G_b, respectively. Consider the coloring

$$g\colon V \to \{1, \ldots, t_r\} \times \{1, \ldots, t_b\}, \text{ where } g(v) = \big(f_r(v), f_b(v)\big).$$

This is a $t_r \cdot t_b$ coloring of G. We show that this coloring is proper, which will imply the desired claim $t_r \cdot t_b \geqslant \chi(G)$. Fix any edge $uv \in E(G)$, and consider the two possible cases:
Case 1: uv is red. Then $f_r(u) \neq f_r(v)$, because f_r is a proper coloring. Thus also $g(u) \neq g(v)$.
Case 2: uv is blue. Then $f_b(u) \neq f_b(v)$, because f_b is a proper coloring. Thus also $g(u) \neq g(v)$. □

Problem 3 (1pt)**.** For every pair of positive integers m, n, construct a graph G with the following properties:

- $|V(G)| = m \cdot n$, and
- $\chi(G) = m$, and
- $\chi(\overline{G}) = n$.

Note: this shows that the equality can be achieved in $\chi(G) \cdot \chi(\overline{G}) \geqslant |V(G)|$.

Solution. Let $G = K_{\underbrace{n,\dots,n}_{m}}$, the complete m-partite graph with parts of size n. Thus $|V(G)| = m \cdot n$.

Since G is m-partite, we have $\chi(G) \leqslant m$. Also, G contains a copy of K_m (pick one vertex for each of the m parts), which shows that $\chi(G) \geqslant m$. Thus $\chi(G) = m$.

Now, $\overline{G}$ is just a disjoint union of m copies of K_n. Since $\chi(K_n) = n$, we have $\chi(\overline{G}) = n$. □

Problem 4 (2pts)**.** Prove that G is 3-critical if and only if $G \cong C_{2n+1}$ for some positive integer n.

Solution. ($\Rightarrow$) Since $\chi(G) = 3$, we know that G is not bipartite. Thus, by Theorem 1.12, G must contain a copy of C_{2n+1} for some positive integer n; let H denote such a subgraph which is isomorphic to C_{2n+1}. If $G \neq H$, then H is a proper subgraph of G. But since $H \cong C_{2n+1}$, we have $\chi(H) = 3$; contradicting the fact that G is 3-critical. Thus $G = H \cong C_{2n+1}$, as needed.

($\Leftarrow$) We already know that $\chi(C_{2n+1}) = 3$. Furthermore, if H is any proper subgraph of C_{2n+1}, then H cannot contain *any* cycle, let alone an odd one (the only cycle in C_{2n+1} is itself). Thus, H is bipartite and so $\chi(H) \leqslant 2$. We conclude that C_{2n+1} is 3-critical. □

Problem 5 (0.5 + 1.5 pts)**.** Recall that, for a graph G with $V(G) = \{v_1, \dots, v_n\}$, the **Mycielski graph** $\mu(G)$ is obtained from G by adding:

(1) *shadow vertices* $\{u_1, \dots, u_n\}$ and edges $u_i v_j \in E(\mu(G))$ if $v_i v_j \in E(G)$,

(2) a vertex w and edges $w u_i$ for all shadow vertices u_i.

Show that:

(a) If G is triangle-free, then $\mu(G)$ is triangle free.

(b) $\chi(\mu(G)) = \chi(G) + 1$.

Solution. (a) Suppose X is a set of three vertices in $\mu(G)$ which form a triangle. Note the following:

- X contains at most one shadow vertex, because no two shadow vertices are adjacent in $\mu(G)$.
- Since w is only adjacent to shadow vertices, it follows that $w \notin X$.
- Also, since G is triangle-free, at least one vertex of X is not in $V(G)$.

It follows that $X = \{v, v', u_i\}$, where $v \neq v' \in V(G)$ are adjacent vertices in G and u_i is a shadow vertex which is a neighbor of v and also of v' in $\mu(G)$. By definition, this means that both v and v' are adjacent to the corresponding vertex v_i in G, contradicting that G is triangle free.

(b) Let $t = \chi G$. We show first that $\chi(\mu(G)) \leqslant t+1$. Consider a proper coloring $f\colon V(G) \to \{1, \dots, t\}$ for G. We extend f to a $(t+1)$-coloring $g\colon V(\mu(G)) \to \{1, \dots, t+1\}$ of $\mu(G)$ by defining

$$g(v_i) = f(v_i), \qquad g(u_i) = f(v_i), \qquad g(w) = t + 1.$$

This coloring is proper, because f is proper and we are using a new color for w. Thus $\chi(\mu(G)) \leqslant t+1$.

Next we show that $\chi(\mu(G)) \geqslant t + 1$. Assume for a contradiction that $\chi(\mu(G)) \leqslant t$, which means that there is a proper coloring $g\colon V(\mu(G)) \to \{1, \dots, t\}$ for $\mu(G)$. By renaming the colors, we can assume that $g(w) = t$. Since w is adjacent to all the shadow vertices, the proper coloring g only uses the colors $\{1, \dots, t-1\}$ for the vertices $\{u_1, \dots, u_n\}$. Define

$$f\colon V(G) \to \{1, \dots, t-1\}, \text{ where } f(v_i) = \begin{cases} g(v_i), & \text{if } g(v_i) \neq t \\ g(u_i), & \text{if } g(v_i) = t \end{cases}$$

(In other words, recolor each vertex in G that has the same g-color as w, using the color of its shadow.) We show that $f(v_i) \neq f(v_j)$ whenever v_i and v_j adjacent vertices in $V(G)$. Since v_i and v_j are also neighbors in $\mu(G)$, we have $g(v_i) \neq g(v_j)$ and up to interchanging v_i and v_j there are two cases:
Case 1: $g(v_i) \neq t$, $g(v_j) = t$. Here $f(v_i) = g(v_i)$ and $f(v_j) = g(u_j)$, But since $v_iv_j \in E(G)$ we have $v_iu_j \in E(\mu(G))$, hence $g(v_i) \neq g(u_j)$, i.e. $f(v_i) \neq f(v_j)$.
Case 2: $g(v_i) \neq t$, $g(v_j) \neq t$. Here $f(v_i) = g(v_i)$ and $f(v_j) = g(v_j)$. But since $v_iv_j \in E(\mu(G))$, we have $g(v_i) \neq g(v_j)$, i.e. $f(v_i) \neq f(v_j)$.

Thus f is a $(t-1)$-proper coloring of G. This contradicts that $\chi(G) = t$ and hence yields the desired result. □

Problem 6 (2pts)**.** Let T be any tree on t vertices, let G be any graph on n vertices and let m be a positive integer. Show that if $n \geqslant (t-1)(m-1)+1$, then either G contains a copy of T or $\overline{G}$ contains a copy of K_m (or both).

Solution. If $m = 1$, then the claim is trivial since every graph contains a copy of K_1; thus we may suppose that $m \geqslant 2$.

Suppose that $\overline{G}$ does not contain a copy of K_m; we need to show that G contains a copy of T. Since $\overline{G}$ does not contain a copy of K_m, we know that $\omega(\overline{G}) \leqslant m-1$, thus $\frac{1}{\omega(\overline{G})} \geqslant \frac{1}{m-1}$. Then, since $\alpha(G) = \omega(\overline{G})$, we have

$$\chi(G) \geqslant \frac{n}{\alpha(G)} = \frac{n}{\omega(\overline{G})} \geqslant \frac{(t-1)(m-1)+1}{m-1} = t-1+\frac{1}{m-1} > t-1.$$

Since $\chi(G)$ and $t-1$ are integers, this means $\chi(G) \geqslant t$. Hence, we know from class that G contains a copy of T. □